SHENG LI

shl188@pitt.edu Homepage LinkedIn Google Scholar

Education

University of Pittsburgh, Pittsburgh, PA, USA 08/2021 – Present
Ph.D. in Computer Science
(Field: Efficient Machine Learning, GenAI, Image/Video Generation, LLM, Self-Supervised Learning)
Advisor: Dr. Xulong Tang

Sichuan University, Chengdu, Sichuan, China 09/2016 – 06/2020
B.Eng. in Software Engineering, Graduated with Distinction

Experience

Apple Inc. 05/2026 - 08/2026
Research Intern *Mentor: Dr. Zhanghao Sun, Dr. Tianyi Zhang*

- **Mixture-of-Attention for Image/Video Generation:** Developed a mixture-of-attention design to adaptively route inputs with different complexity to different types of attention mechanisms, reducing over **80%** attention costs for image/video generation without affecting the generation quality.

Adobe Research 05/2025 – 11/2025
Research Scientist/Engineer Intern *Mentor: Dr. Yifan Gong*

- **Dynamic Patchification for Video Diffusion:** Developed a content-aware patchification router that adaptively adjusts patch size (i.e., process granularity) across different spatiotemporal regions of video, reducing redundant tokens and achieving **1.8× speedup** for video generation without affecting visual quality.

University of Pittsburgh 2020 – Present
Research Assistant / Teaching Assistant *Advisor: Dr. Xulong Tang*

- **Video Diffusion Pruning:** Proposed a training-free temporal-aware token pruning method for video generation that preserves temporal coherence during pruning and achieves **1.5× speedup** without quality loss.
- **Efficient Self-Supervised Learning (SSL):** Developed a similarity-based pruning method to discard less important input image regions in discriminative SSL, cutting training costs by over **40%**.
- **Efficient LLM Serving:** Accelerated RAG-enabled large language model serving by deduplicating and reordering retrieved RAG chunks and improving KV-cache reuse, reducing redundant chunk transfers by $\sim$**25%** and end-to-end latency by $\sim$**18%** (Llama-3.1-8B on vLLM).
- **Layer Freezing for Training:** Proposed an attention-based layer freezing approach that reduces computation by **40%** and memory cost by **50%** for ML model training.
- **3DGS Acceleration:** Leveraged tensor cores to accelerate the 3D Gaussian Splatting rendering.

Selected Publications *(*equal contribution)*

[1] **Sheng Li**, Connelly Barnes, Mamshad Nayeem Rizve, Hongwu Peng, Zhengang Li, Ohi Dibua, Alireza Ganjdanesh, Xulong Tang, Yan Kang, and Yifan Gong. "Content-Aware Dynamic Patchification for Efficient Video Diffusion." *In Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026.* **(CVPR 2026).**

[2] **Sheng Li***, Qitao Tan*, Yue Dai, Zhenglun Kong, Tianyu Wang, Jun Liu, Ao Li, Ninghao Liu, Yufei Ding, Xulong Tang, and Geng Yuan. "Mutual Effort for Efficiency: A Similarity-based Token Pruning for Vision Transformers in Self-Supervised Learning." *In Proceedings of 13th International Conference on Learning Representations.* **(ICLR 2025)**

[3] **Sheng Li**, Chao Wu, Ao Li, Yanzhi Wang, Xulong Tang, and Geng Yuan. "Waxing-and-Waning: a Generic Similarity-based Framework for Efficient Self-Supervised Learning." *In Proceedings of 12th International Conference on Learning Representations.* **(ICLR 2024)**

[4] **Sheng Li***, Geng Yuan*, Yue Dai*, Youtao Zhang, Yanzhi Wang, and Xulong Tang. "SmartFRZ: An Efficient Training Framework using Attention-Based Layer Freezing." *In Proceedings of 11th International Conference on Learning Representations.* **(ICLR 2023, Spotlight Paper)**

[5] **Sheng Li**, Yang Sui, Junhao Ran, Bo Yuan, Yue Dai, and Xulong Tang. "Temporal Aware Pruning for Efficient Diffusion-based Video Generation." **(Preprint)**

[6] **Sheng Li**, Yang Sui, Yue Wu, Zhuoran Song, Bo Yuan, Xulong Tang, and Yue Dai. "Accelerating 3D Gaussian Splatting using Tensor Cores." **(Preprint)**

[7] **Sheng Li**, Geng Yuan, Yawen Wu, Yue Dai, Tianyu Wang, Chao Wu, Alex K. Jones, Jingtong Hu, Yanzhi Wang, and Xulong Tang. "ETuner: A Redundancy-Aware Framework for Efficient Continual Learning Application on Edge Devices." **(Preprint)**

[8] Tianyu Wang*, **Sheng Li***, Bingyao Li, Yue Dai, Ao Li, Geng Yuan, Yufei Ding, Youtao Zhang, and Xulong Tang. "Improving GPU Multi-Tenancy Through Dynamic Multi-Instance GPU Reconfiguration." **(Preprint)**

Selected Honors and Awards

Spotlight Award, The 11th ICLR Conference	2023
National Scholarship, Ministry of Education of China	2018, 2019
Comprehensive Scholarship, Sichuan University	2017

Skills

Python, PyTorch; Generative AI, Diffusion, Video Generation; LLM Serving (RAG, KV-cache reuse); Computer Vision, Self-Supervised Learning; Efficient ML Methods (Token Pruning, Layer Freezing); CNN, Transformer; GPU Programming (CUDA), GPU Architecture, Nsight GPU Profiling, ML Kernel; 3DGS

Teaching

Teaching Assistant: Computer Organization and Assembly Language; Introduction to Systems Software; Algorithms and Data Structures

Professional Service

Reviewer of NeurIPS, ICLR, ICML, CVPR, ECCV, AAAI, TPAMI, TICPS, ACML, and NPL